

LAN (Local Area Network): LAN is a network that connects computers and devices in a limited geographic area, such as a home, office, or campus. It allows for the sharing of resources like files and printers.

WAN (Wide Area Network): WAN is a network that spans a larger geographic area, often connecting LANs across cities or even continents. The internet is a vast example of a global WAN.

Fibre Optic: Fiber optic cables use light signals to transmit data at high speeds. They are known for their high bandwidth and are commonly used for long-distance and high-speed internet connections.

Twisted Pair Cable: Twisted pair cables are common for Ethernet connections. They consist of pairs of insulated copper wires twisted together to reduce electromagnetic interference.

Modem (Modulator-Demodulator): A modem converts digital data from a computer into analog signals for transmission over telephone lines (DSL) or cable lines (cable modem). It also converts incoming analog signals back into digital data.

Switch: A network switch connects devices within a LAN, allowing them to communicate efficiently. It operates at the data link layer (Layer 2) of the OSI model.

Router: A router connects different networks, such as a home network to the internet. It forwards data packets between networks based on their IP addresses.

Firewall: A firewall is a security device or software that monitors and controls incoming and outgoing network traffic. It helps protect a network from unauthorized access and cyber threats.

VSAT (Very Small Aperture Terminal): VSAT is a satellite communication system that enables remote locations to access the internet or connect to corporate networks via small satellite dishes.

Submarine Cable: Submarine cables are fiber optic cables laid on the ocean floor to connect continents and facilitate international data transmission.

SEA-ME-WE (South-East Asia-Middle East-Western Europe): SEA-ME-WE is a consortium of submarine cables that provides high-capacity internet connectivity between regions in Asia, the Middle East, and Western Europe.

BTCL (Bangladesh Telecommunications Company Limited): BTCL is the state-owned telecommunications company in Bangladesh, responsible for providing various telecommunications services, including internet connectivity.

Mobile Data: Mobile data refers to internet access provided by cellular networks, allowing users to connect to the internet via smartphones, tablets, or mobile hotspots.

BTS (Base Transceiver Station): BTS is a key component of cellular networks. It connects mobile devices to the network, facilitating voice and data communication.

MC (Mobile Carrier): A mobile carrier is a telecommunications company that provides cellular services to consumers, including voice calls, text messaging, and mobile data. Examples include AT&T, Verizon, and Vodafone.

These terms represent various components and technologies that together form the infrastructure of the internet and telecommunications networks, enabling global connectivity and data transmission.

Operating System (OS):

History: Operating systems have evolved over time. Early computers didn't have operating systems, and programs were loaded directly. In the 1950s and 1960s, systems like UNIVAC I and IBM OS/360 introduced basic OS functions. In the 1970s, Unix revolutionized OS design, leading to modern OSs like Windows, macOS, and Linux.

Mythical Man-Month: "The Mythical Man-Month" is a book by Fred Brooks that discusses software project management. It emphasizes that adding more people to a late project only makes it later, highlighting the complexities of coordinating large teams.

Resident Monitor: Early computers used resident monitors, basic OS components stored in memory permanently. They provided limited functionality and required frequent manual intervention.

Kernel: The kernel is the core of an OS, managing hardware resources and providing essential services. It controls memory, processes, file systems, and device drivers.

Micro Kernels: Microkernel architecture separates the kernel into small, modular components, each running in user mode. This design aims for simplicity, reliability, and extensibility.

Monolithic Kernel: In a monolithic kernel, the entire OS runs as a single program in kernel mode. While efficient, it's less modular and robust than microkernels.

Unix File Structure and File Protection:

Unix File Structure: Unix uses a hierarchical file system, starting with the root directory ("/"). Directories contain files and subdirectories. Paths are expressed as "/directory/filename."

File Protection: Unix employs a permission system with three levels: user, group, and others. Permissions include read (r), write (w), and execute (x). Users can set file permissions to control access.

Linux Commands:

Linux commands are used to interact with the Linux operating system via a terminal. Common commands include:

ls: List files and directories.

cd: Change the current directory.

pwd: Print the current working directory.

mkdir: Create a new directory.

rm: Remove files or directories.

cp: Copy files or directories.

mv: Move or rename files and directories.

touch: Create an empty file.

cat: Concatenate and display file content.

grep: Search for text in files.

chmod: Change file permissions.

chown: Change file ownership.

ps: List running processes.

kill: Terminate processes.

ifconfig: Configure network interfaces.

ping: Test network connectivity.

apt or yum: Package managers for software installation.

These commands allow users to manage files, navigate the file system, control processes, and configure system settings in Linux-based operating systems. Learning these commands is fundamental for Linux system administration and daily usage.

here's a brief description of each of the mentioned I/O (Input/Output) devices:

Keyboard:

A keyboard is a common input device that allows users to enter alphanumeric characters, symbols, and commands into a computer or other digital devices. It typically consists of a set of keys arranged in a QWERTY layout.

Mouse:

A mouse is an input device used to move a cursor on a computer screen and interact with graphical user interfaces. It typically has two buttons and a scroll wheel, enabling users to click, double-click, right-click,

and scroll.

CRT Monitor (Cathode Ray Tube Monitor):

A CRT monitor is an older type of display technology that uses cathode ray tubes to display images on a screen. They were once common but have largely been replaced by newer display technologies like LCD and LED due to their bulk and energy consumption.

LCD Display (Liquid Crystal Display):

An LCD display is a flat-panel display technology that uses liquid crystals to modulate the passage of light and display images. LCDs are commonly used in laptops, computer monitors, and various handheld devices.

LED Display (Light-Emitting Diode Display):

An LED display is a flat-panel display technology that uses light-emitting diodes to produce images. LED displays are known for their energy efficiency, brightness, and vibrant colors. They are used in TVs, digital billboards, and various electronic devices.

These I/O devices are essential components of a computer system, allowing users to input data (keyboard and mouse) and receive visual output (monitors and displays). They play a crucial role in facilitating user interaction with digital devices and computers.

Pseudocode for Testing Prime Numbers (Basic Version):

plaintext

Copy code

Function isPrime(n)

 if n is less than 2

 return false

 for i from 2 to the square root of n

 if n is divisible by i

```
    return false
```

```
    return true
```

This pseudocode outlines a basic algorithm to check if a number n is prime. It starts with a simple optimization by checking if n is less than 2 (prime numbers are greater than 1). Then, it iterates from 2 up to the square root of n and checks if any number divides n evenly. If it finds such a divisor, it returns false; otherwise, it returns true.

Improved Pseudocode for Testing Prime Numbers (Optimized Version):

plaintext

Copy code

Function isPrime(n)

```
    if  $n$  is less than 2
```

```
        return false
```

```
    if  $n$  is 2
```

```
        return true
```

```
    if  $n$  is even
```

```
        return false
```

```
    for  $i$  from 3 to the square root of  $n$ , incrementing by 2
```

```
        if  $n$  is divisible by  $i$ 
```

```
            return false
```

```
    return true
```

In this improved pseudocode, we've added more optimizations:

We check if n is 2 (the only even prime) and return true immediately. This avoids unnecessary iteration for even numbers greater than 2.

We check if n is even (except for 2) and return false immediately, as all other even numbers cannot be

prime.

We start the loop at 3 and increment by 2 in each step. This avoids testing even numbers in the loop because we've already excluded them.

These optimizations reduce the number of iterations, making the algorithm more efficient. However, the basic algorithm and the improved version both have a time complexity of $O(\sqrt{n})$, where n is the number being tested. This is the most efficient known way to determine if a number is prime without employing more advanced mathematical techniques.

here are brief descriptions of terms related to web architecture:

DHCP (Dynamic Host Configuration Protocol):

DHCP is a network protocol that automatically assigns IP addresses, subnet masks, and other network configuration parameters to devices within a local network. It simplifies network management by dynamically allocating addresses.

ARP (Address Resolution Protocol):

ARP is a protocol used to map an IP address to a physical MAC (Media Access Control) address on a local network. It helps devices communicate within the same network segment.

RARP (Reverse ARP):

RARP is an older protocol that converts a MAC address into an IP address. It was primarily used for diskless workstations to obtain their IP addresses.

BOOTP (Bootstrap Protocol):

BOOTP is a precursor to DHCP and is used to automatically configure network settings for devices, typically in the booting process. DHCP evolved from BOOTP to offer more advanced features.

DNS (Domain Name System):

DNS is a system that translates human-readable domain names (e.g., www.example.com) into IP addresses that computers use to identify each other on the internet. It facilitates easy access to websites and resources.

IP (Internet Protocol):

IP is a fundamental protocol of the internet that enables data packets to be routed between devices in different networks. IPv4 and IPv6 are two versions of IP in use today.

Port:

A port is a numeric identifier used to distinguish between different services or processes on a single device. Ports allow multiple network services to run on the same device without conflict.

Socket:

A socket is a combination of an IP address and a port number. It enables two-way communication between devices over a network, often used in network programming.

URL (Uniform Resource Locator):

A URL is a web address that specifies the protocol (e.g., HTTP), domain name (e.g., www.example.com), and path to access a resource on the internet.

HTTP Protocol (Hypertext Transfer Protocol):

HTTP is a protocol used for transferring hypertext, such as web pages and multimedia content, between a web server and a client (usually a web browser).

GET/POST (HTTP Request Methods):

GET and POST are HTTP request methods. GET is used to retrieve data from a server, while POST is used to send data to be processed by a server, often used for submitting forms on websites.

Request/Response (HTTP Communication):

In HTTP, communication occurs through a request-response model. A client (e.g., web browser) sends an HTTP request to a server, and the server responds with the requested content or an error message.

Stateless Protocol:

HTTP is a stateless protocol, meaning each request from a client to a server is independent. The server does not retain information about previous requests, requiring mechanisms like cookies or sessions to maintain state.

Session (Web Session):

A session is a way to track a user's interactions with a website over multiple requests. It allows for the persistence of user-specific data, such as login status, across different pages of a site.

These terms are fundamental to understanding the architecture and operation of the World Wide Web and internet communication protocols.